

Price-to-earnings ratios not only are used for individual stocks but also are a valuation metric for stock indexes, such as S&P 500, FTSE 100, or Nikkei 225. Here, the stock index value is divided by a weighted sum of the index constituents' earnings per share. This will be explored in depth later in the curriculum, but we can relate the price-to-earnings ratio to our earlier discussion of relating a stock's price (PV) to expected future cash flows to make some useful observations. First, recall Equation 14:

$$PV_t = \frac{D_t(1+g)}{r-g}.$$

We can divide both sides by E_t , earnings per share for period t , to obtain:

$$\frac{PV_t}{E_t} = \frac{\frac{D_t}{E_t} \times (1+g)}{r-g}. \quad (23)$$

The left-hand side of Equation 23 is the price-to-earnings ratio, whereas the first term in the numerator on the right is the proportion of earnings distributed to shareholders as dividends known as the **dividend payout ratio**.

Given a price-to-earnings ratio and dividend payout ratio, we can solve for either required return or implied dividend growth rate (given an assumption about the other). The required return is useful in understanding investor return expectations on a forward-looking basis. The implied constant growth rate is useful to compare with the company's expected growth rate and historical growth rate. For example, if the implied constant growth rate is 10 percent yet the analyst estimates that the company can only grow by 5 percent, the analyst may judge the shares to be overvalued.

In practice, the **forward price-to-earnings ratio** or ratio of its share price to an estimate of its next period ($t+1$) earnings per share is commonly used. With it, we can simplify the previous equation as follows:

$$\frac{PV_t}{E_{t+1}} = \frac{\frac{D_{t+1}}{E_{t+1}}}{r-g}. \quad (24)$$

From Equation 24, we can see that forward price-to-earnings ratio is positively related to higher expected dividend payout ratio and higher expected growth but is negatively related to the required return.

EXAMPLE 11

Implied Return and Growth from Price to Earnings Ratio

1. Suppose Coca-Cola stock trades at a forward price to earnings ratio of 28, its expected dividend payout ratio is 70 percent, and analysts believe that its dividend will grow at a constant rate of 4 percent per year. Calculate Coca-Cola's required return.

Solution:

6.50 percent

Using Equation 24,

$$28 = \frac{0.7}{r - 0.04}.$$

Solving this equation for r , Coca-Cola's required return is 6.50 percent.

Given the above result for Coca-Cola's required return, it should come as no surprise that if we instead assume that the required return on Coca-Cola stock is 6.50 percent, then we would find that Coca-Cola's implied growth rate is 4 percent, and this result is confirmed in question 2.